

Corollary 10 (proof in Appendix E.10) *We assume $\alpha \in (0, e^{-1})$, $\beta \in (0, 1)$, $s > 0$, $R > 0$, $M > 0$, $m + n > 15$ so that $\ln(\ln(m + n)) > 1$ and $B_1, B_2, B_3 \in \mathbb{N}$ satisfying $B_1 \geq \frac{3}{\alpha^2}(\ln(\frac{8}{\beta}) + \alpha(1 - \alpha))$, $B_2 \geq \frac{8}{\alpha^2} \ln(\frac{2}{\beta})$ and $B_3 \geq \log_2(\frac{2\pi^2}{3\alpha})$. We consider our aggregated MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ with the collection of bandwidths*

$$\Lambda := \left\{ (2^{-\ell}, \dots, 2^{-\ell}) \in (0, \infty)^d : \ell \in \left\{ 1, \dots, \left\lceil \frac{2}{d} \log_2 \left(\frac{m + n}{\ln(\ln(m + n))} \right) \right\rceil \right\} \right\}$$

and the collection of positive weights $w_\lambda := \frac{6}{\pi^2 \ell^2}$ so that $\sum_{\lambda \in \Lambda} w_\lambda \leq 1$ for any sample sizes m and n . The uniform separation rate of the MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ over the Sobolev ball $\mathcal{S}_d^s(R)$ then satisfies

$$\rho(\Delta_\alpha^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M) \leq C_7(M, d, s, R, \alpha, \beta) \left(\frac{m + n}{\ln(\ln(m + n))} \right)^{-2s/(4s+d)}$$

for some positive constant $C_7(M, d, s, R, \alpha, \beta)$. This means that the MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$, which does not depend on s and R , is optimal in the minimax sense up to an iterated logarithmic term over the Sobolev balls $\{\mathcal{S}_d^s(R) : s > 0, R > 0\}$; the MMDAgg test $\Delta_\alpha^{\Lambda^w, B_{1:3}}$ is minimax adaptive.

Note that the choice of using negative powers of 2 for the bandwidths in Corollary 10 is arbitrary. The result holds more generally using negative powers of a for any real number $a > 1$.

With the specific choice of bandwidths and weights of Corollary 10, we have proved that the uniform separation rate of the proposed aggregated MMDAgg test is upper bounded by $((m + n)/\ln(\ln(m + n)))^{-2s/(4s+d)}$. Comparing this with the minimax rate $(m + n)^{-2s/(4s+d)}$, we see that MMDAgg attains rate optimality over the Sobolev ball $\mathcal{S}_d^s(R)$, up to an iterated logarithmic factor, and more importantly, the aggregated test does not depend on the prior knowledge of the smoothness parameter s . Our MMDAgg test is minimax adaptive over the Sobolev balls $\{\mathcal{S}_d^s(R) : s > 0, R > 0\}$.

4 Related work

In this section, we compare our results to a number of different adaptive kernel hypothesis testing approaches.

Fromont et al. (2012, 2013) construct a two-sample aggregated test in a framework in which the sample sizes follow independent Poisson processes. They use a different kernel-based estimator which corresponds to an unscaled version of the classical quadratic-time MMD estimator of Gretton et al. (2012a, Lemma 6). In their Poisson setting, they derive uniform separation rates for their aggregated test which is minimax adaptive over Sobolev balls, and over anisotropic Nikol'skii-Besov balls, up to an iterated logarithmic term. The quantiles they consider are estimated with a wild bootstrap. They also have an additional assumption on the kernel (condition in the Fourier domain; Fromont et al., 2013, Equation 3.7), which we do not require.

Albert et al. (2022) consider the problem of testing whether two random vectors are dependent and use the kernel-based Hilbert-Schmidt Independence Criterion (HSIC—Gretton

et al., 2005) as a dependence measure. Similarly to our work, they propose a non-asymptotic minimax adaptive test which aggregates single (HSIC) tests, and provide theoretical guarantees: upper bounds for the uniform separation rate of testing over Sobolev and Nikol'skii balls. In their independence testing setting, the information about the problem is encoded in the joint distribution over pairs of variables, with the goal of determining whether this is equal to the product of the marginals. This differs from the two-sample problem we consider, where we have samples from two separate distributions.

Albert et al. (2022) define their single HSIC test using the theoretical quantile of the statistic under the null hypothesis, which is an unknown quantity in practice. To implement the test, they propose a deterministic upper bound on the theoretical quantile (Albert et al., 2022, Proposition 3). This upper bound holds in the two-sample case under the restrictive assumption $(m+n)\sqrt{\lambda_1 \cdots \lambda_d} > \ln(\frac{1}{\alpha})$ (this condition is adapted to the two-sample setting from their condition $n\sqrt{\lambda_1 \cdots \lambda_p \mu_1 \cdots \mu_q} > \ln(\frac{1}{\alpha})$ for independence testing). If the bandwidth is small (as it can be in the case of the optimal bandwidth λ^* in the proof of Corollary 10), then this condition implies that the results would hold only for very large sample sizes.

By contrast with the above bound, we use a wild bootstrap or permutations to approximate the theoretical quantiles. While the theoretical quantiles are real numbers given data, our estimated quantiles are random variables given data. This means that instead of having a deterministic upper bound on the theoretical quantiles (Albert et al., 2022, Proposition 3), we have an upper bound on our estimated conditional quantiles which holds with high probability as in Proposition 4. Our use of an estimated threshold in place of a deterministic upper bound has an important practical consequence: it allows us to drop the assumption $(m+n)\sqrt{\lambda_1 \cdots \lambda_d} > \ln(\frac{1}{\alpha})$ entirely.

Another difference is how the level correction of the single MMD tests is performed. The aggregated test of Albert et al. (2022) involves a theoretical value u_α which cannot be computed in practice, we incorporate directly in our test a Monte Carlo approximation, using either a wild bootstrap or permutations, to estimate the probability under the null hypothesis, and use the bisection method to approximate the supremum. We stress that our theoretical guarantees of minimax optimality (up to an iterated logarithmic term) hold for our aggregated MMDAgg test which can be implemented without any further approximations. Finally, while the results of Albert et al. (2022) hold only for the Gaussian kernel, ours are more general and hold for any product of one-dimensional characteristic translation invariant kernels which are absolutely and square integrable.

Kim et al. (2022, Section 7) propose an adaptive two-sample test for testing equality between two Hölder densities supported on the real d -dimensional unit ball. Instead of testing various bandwidths or kernels, they discretise the support in bins of equal sizes and aggregate tests with varying bin sizes. Each single test is a multinomial test based on the discretised data. Their strategy and the function class they use are both different from the one we consider, but they derive a similar upper bound on the uniform separation rate of testing over Hölder densities. Kim et al. (2022, Proposition 8.4) also mention the setting considered by Albert et al. (2022) and prove an equivalent version of our Theorem 5 for single tests, using permutations for the Gaussian kernel. We consider both the permutation-based and wild bootstrap procedures, and our results hold more generally for a wide range of kernels. With those aforementioned differences, Kim et al. (2022, Example 8.5) anticipate

that one can use a similar reasoning to Albert et al. (2022) to obtain minimax optimality of the single MMD tests. We provide the full statement and proof of this result in our more general setting.

Li and Yuan (2019) present goodness-of-fit, two-sample and independence aggregated asymptotic tests and also establish the minimax rates over Sobolev balls for these three settings. Their tests use the Gaussian kernel and heavily rely on the asymptotic distribution of the test statistic, while our test is non-asymptotic and is not limited to a particular choice of kernel. Their tests are adaptive over Sobolev balls (which they define in a slightly different way than in our case) provided that the smoothness parameter satisfies $s \geq d/4$. We do not have such a restriction. We also note that they assume that the two densities belong to a Sobolev ball, rather than assuming only that the difference of the densities lies in a Sobolev ball. We also point out the work of Tolstikhin et al. (2016) who derive lower bounds for MMD estimation based on finite samples for any radial universal kernel (Sriperumbudur et al., 2011). They establish the minimax rate optimality of the MMD estimators (V -statistic and U -statistic; Lee, 1990).

Gretton et al. (2012b) address kernel adaptation for the linear-time MMD, where the test statistic is computed as a running average (this results in a statistic with greater variance, but allows the processing of larger sample sizes). They propose to choose the kernel by splitting the data, and using one part to select the bandwidth which maximises the estimated ratio of the Maximum Mean Discrepancy to the standard deviation. They show that maximizing this criterion for the linear-time setting corresponds to maximizing the asymptotic power of the test. The test is then performed on the remaining part of the data. Sutherland et al. (2017) and Liu et al. (2020) address kernel adaptation for the quadratic-time MMD using the same sample-splitting strategy, and show that the ratio of the MMD to the standard deviation under the alternative can again be used as a good proxy for test power. Liu et al. (2020) in particular propose a regularized estimator for the variance under the alternative hypothesis, which admits a convenient closed-form expression. Generally, kernel choice by sample splitting gives better results than the median heuristic, as the former is explicitly tuned to optimize the asymptotic power (or a proxy for it). The price to pay for this increase in performance, however, is that we cannot use all the data for the test. In cases where we have access to almost unlimited data this clearly would not be a problem, but in cases where we have a restricted number of samples and work in the non-asymptotic setting, the loss of data to kernel selection might actually result in a net reduction in power, even after kernel adaptation. For better data efficiency, Kübler et al. (2022a) propose an MMD test which uses held-out data not only for kernel selection, but also for choosing weights and test locations for the MMD witness function. In later work, leveraging recent advances in supervised learning and also relying on sample splitting, Kübler et al. (2022b) construct a test which learns the witness function directly by training, for a given amount of time (*i.e.* one minute), an AutoGluon model (Erickson et al., 2020) which can be, for example, a neural network.

Kübler et al. (2020) propose another approach to an MMD adaptive two-sample test which does not require data splitting. Using all the data, they select the linear combination of test statistics with different bandwidths (or even different kernels) which is optimal in the sense that it maximises a power proxy, they then run their test using again all the data. Using the post-selection inference framework (Fithian et al., 2014; Lee et al., 2016), they